Bi-weekly Report

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Group No: 14
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Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

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Objectives for this report

1. Heat exchanger network design using pinch technology.
2. Estimate utility costs. Perform detailed techno-economic analysis.

1 Basis

Report 4 closed the flowsheet with all four recycles converged. Heat integration and costing were left for this report. The items open at that time are in Table 1.

Table 1: Position at the close of Report 4.

Item	Report 4 status	Taken up in
Pinch analysis of the full network	Not done	Section 2
Reactor heat, 15.93 MW	No utility assigned	Section 3
COOL-1 duty, 4.14 MW	No utility assigned	Section 3
Utility costs	Not estimated	Section 4
Capital cost	Not estimated	Section 5
NPV, IRR and payback	Not estimated	Section 6
Aniline purity, 99.9 wt% required	99.45 wt%	Section 7
Nitrobenzene in product, below 5 ppm	313 ppm	Section 7
Aniline in wastewater, below 1 ppm	216 ppm into carbon bed	Section 7
Cyclohexylamine in wastewater	4.28 wt%	Sections 6, 7

The flowsheet is the same as in Report 4. A copy of che453_R4_FINAL is saved as che453_R5_00.apwz and the Energy Analysis of Aspen Plus V14 is run on it. The stream data is then taken into Aspen Energy Analyzer. The targets and the network are in

che453_R5_01_Copy1.hch, design A_Design2. The final Aspen Plus file is che453_R5_01.apwz. All costs are in US dollars at the 2023 annual CEPCI of 797.9, and Rs. 95 per USD is used as in Report 1.

1.1 Key results

On the Aspen Energy Analyzer data the plant is a threshold problem. At $\Delta T_{min} = 10$ K the minimum hot utility is zero and the minimum cold utility is 21.33 MW. The reactor alone gives 15.93 MW at 300°C, which is hot enough for every heater and reboiler, but it covers only part of the 22.24 MW demand.

Aspen Energy Analyzer gives a network of 17 exchangers which meets this target exactly. In the plant the reactor coolant cannot be as hot as Aspen assumes, because of the bed temperature limit of Report 4. So the final network has the reactor shell raising 40 bar steam, with three new exchangers. Purchased heating falls from 22.24 MW to 1.80 MW. Cooling water falls from 3137 t/h to 1327 t/h, taking REACT and COOL-1 on cooling water in the base case. The 1.80 MW left is the vaporisation of nitrobenzene at 262°C, which is above the 40 bar steam temperature. Utility cost falls from USD 10.43 million per year to USD 0.67 million per year.

The fixed capital is USD 18.2 million (Rs. 173 crore) and the total capital USD 21.0 million (Rs. 199 crore). At Report 1 prices the project gives an NPV of USD 61.3 million at 12 per cent, an IRR of 50.4 per cent, a return on investment of 55 per cent and a payback of 1.6 years. Without the heat integration the NPV is only USD 16.4 million.

2 Pinch Analysis

2.1 Stream data

The stream data was taken by the Energy Analysis of Aspen Plus straight from the converged flowsheet. Every duty was checked against the utility and model workbooks of R4_05 and agrees to the third decimal. Three corrections were needed before the targets could be used.

- REACT was read in as a cold stream from 160.2 to 300°C, heated by fired heat. This is because the reactor outlet is hotter than its inlet. In reality the bed gives off 15.93 MW at 300°C. It was deleted and re-entered as a hot stream from 301 to 300°C with the same heat load.
- DEC and FLASH were read as small duties at 40°C, 0.18 MW together. The 0.18 MW at 40°C is small and near ambient, so it was left out.
- VAP-1 starts at 28.0°C and not at 144°C. The 144 is the molar flow of S1 in kmol/h. The duty of 3.68 MW is unchanged.

The final data is in Table 2. VAP-1 and the FEHE cold side make one stream, S1 to S3. COOL-1, COOL-2 and the FEHE hot side make one stream, RXOUT to S6.

Table 2: Stream data used for pinch analysis, from Aspen Energy Analyzer.

Stream	Type	T_{in} (°C)	T_{out} (°C)	Duty (MW)
REACTOR (bed, re-entered)	Hot	301.0	300.0	15.93
RXOUT to S6 (FEHE, COOL-1, COOL-2)	Hot	300.0	40.0	9.58
ANRECOV to ARLIQ (ARCOND)	Hot	97.2	40.0	7.63
COL-2 condenser	Hot	131.6	129.0	7.02
COL-1 condenser	Hot	101.8	64.7	2.93
PRODWARM to PRODUCT (PRODCOOL)	Hot	129.0	40.0	0.66
S1 to S3 (VAP-1, FEHE)	Cold	28.0	290.0	3.89
COL-1 reboiler	Cold	176.1	185.8	4.06
COL-2 reboiler	Cold	143.9	145.4	6.56
STRIP reboiler	Cold	100.9	101.2	7.94
Total hot / total cold				43.78 / 22.45

2.2 Manual Problem Table Algorithm (PTA)

To independently verify the energy targets, the Problem Table Algorithm (PTA) is performed manually on the corrected stream data using a minimum approach temperature $\Delta T_{min} = 10$ K.

Step 1: Shift Temperatures We first calculate the shifted temperatures T^* for all hot and cold streams. For hot streams: $T^* = T_{hot} - \frac{\Delta T_{min}}{2} = T_{hot} - 5°\text{C}$ For cold streams: $T^* = T_{cold} + \frac{\Delta T_{min}}{2} = T_{cold} + 5°\text{C}$

The heat capacity flow rate ($CP = \frac{\Delta H}{\Delta T}$) is also calculated for each stream.

Table 3: Shifted Temperatures and Heat Capacity Flow Rates

Stream	Type	Duty (MW)	T_{in} (°C)	T_{out} (°C)	Shifted T^*_{in}	Shifted T^*_{out}	CP (MW/K)
REACTOR	Hot	15.93	301.0	300.0	296.0	295.0	15.9300
RXOUT to S6	Hot	9.58	300.0	40.0	295.0	35.0	0.0368
ARCOND	Hot	7.63	97.2	40.0	92.2	35.0	0.1334
COL-2 cond	Hot	7.02	131.6	129.0	126.6	124.0	2.7000
COL-1 cond	Hot	2.93	101.8	64.7	96.8	59.7	0.0790
PRODCOOL	Hot	0.66	129.0	40.0	124.0	35.0	0.0074
S1 to S3	Cold	3.89	28.0	290.0	33.0	295.0	0.0148
COL-1 reb	Cold	4.06	176.1	185.8	181.1	190.8	0.4186
COL-2 reb	Cold	6.56	143.9	145.4	148.9	150.4	4.3733
STRIP reb	Cold	7.94	100.9	101.2	105.9	106.2	26.4667

Step 2: Define Temperature Intervals Sorting the shifted temperatures in descending order creates the temperature intervals. The sum of $CP_{hot} - CP_{cold}$ is calculated for each interval.

Sorted T^* values (°C): 296.0, 295.0, 190.8, 181.1, 150.4, 148.9, 126.6, 124.0, 106.2, 105.9, 96.8, 92.2, 59.7, 35.0, 33.0.

Step 3: Heat Cascade Calculation We cascade the heat down the temperature intervals, starting with an assumed hot utility $Q_H = 0$.

Table 4: Heat Cascade Calculation

T^* (°C)	Active Streams	$\sum CP_H - \sum CP_C$	ΔT	ΔH_{int} (MW)
296.0				
	REACTOR	15.9300	1.0	15.930
295.0				
	RXOUT, S1-S3	0.0220	104.2	2.292
190.8				
	RXOUT, S1-S3, COL-1 reb	-0.3966	9.7	-3.847
181.1				
	RXOUT, S1-S3	0.0220	30.7	0.675
150.4				
	RXOUT, S1-S3, COL-2 reb	-4.3513	1.5	-6.527
148.9				
	RXOUT, S1-S3	0.0220	22.3	0.491
126.6				
	RXOUT, S1-S3, COL-2 cond	2.7220	2.6	7.077
124.0				
	RXOUT, S1-S3, PRODCOOL	0.0294	17.8	0.523
106.2				
	RXOUT, S1-S3, PRODCOOL, STRIP	-26.4373	0.3	-7.931
105.9				
	RXOUT, S1-S3, PRODCOOL	0.0294	9.1	0.268
96.8				
	RXOUT, S1-S3, PRODCOOL, COL-1 cond	0.1084	4.6	0.499
92.2				
	RXOUT, S1-S3, PRODCOOL, COL-1 cond, ARCOND	0.2418	32.5	7.859
59.7				
	RXOUT, S1-S3, PRODCOOL, ARCOND	0.1628	24.7	4.021
35.0				
	S1-S3	-0.0148	2.0	-0.030
33.0				

Step 4: Target Identification Since the lowest value in the cascade is 0.000 (occurring at the highest temperature level), the initial assumption of $Q_H = 0$ is correct. There is no process pinch point.

The targets at $\Delta T_{min} = 10$ K are:

- **Minimum Hot Utility ($Q_{H,min}$) = 0 MW**
- **Minimum Cold Utility ($Q_{C,min}$) = 21.30 MW** (Note: Small difference from 21.33 MW due to rounding in table).

This manual calculation confirms that this is a threshold problem and matches the results gen-

erated by Aspen Energy Analyzer.

2.3 Targets

The composite curves are in Figure 1 and the grand composite curve in Figure 2. The hot curve is fully above the cold curve. So there is no process pinch and this is a threshold problem. When MP steam raising is added as a utility, Aspen puts a utility pinch at 153.9°C hot and 143.9°C cold, just above the COL-2 reboiler. The targets are in Table 5.

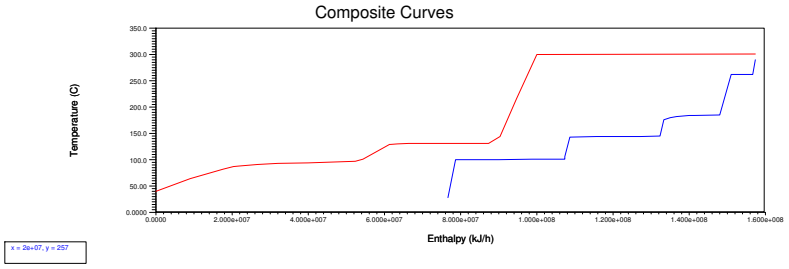


Figure 1: Composite curves at $\Delta T_{min} = 10$ K, from Aspen Energy Analyzer. Red is the hot composite and blue the cold composite.

Table 5: Energy targets at $\Delta T_{min} = 10$ K, Aspen Energy Analyzer.

Item	Value
Minimum hot utility	0 MW
Minimum cold utility	21.33 MW
Current heating in flowsheet	22.24 MW
Number of units, minimum / at maximum energy recovery	11 / 12
Number of shells	18
Area target, counter-current / 1-2 shell and tube	4389 / 4445 m^2
MP steam that can be raised, at 175°C	4.70 MW
Cooling water left, with steam raised	16.63 MW
Utility pinch, hot / cold	153.9 / 143.9°C

The 4.70 MW MP steam target includes Aspen's default utility settings for HP/MP splitting and BFW preheat. The GCC shows a large pocket from 300 down to about 100°C. Inside it the reactor heat runs down into the reboilers. Below 100°C all the heat goes to cooling water.

Table 6 gives the effect of ΔT_{min}, from a problem table on the same data. The hot utility stays

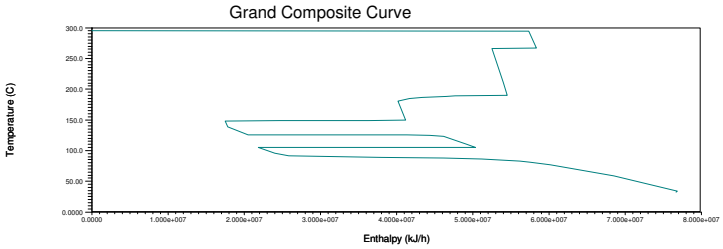


Figure 2: Grand composite curve at $\Delta T_{min} = 10$ K, from Aspen Energy Analyzer.

near zero up to 30 K. At 40 K it jumps to 1.83 MW, because the reactor at 300°C can no longer vaporise the feed at 262°C. This value comes back in Section 3.

Table 6: Effect of ΔT_{min} on the targets.

ΔT_{min} (K)	Hot utility (MW)	Cold utility (MW)
5	0	21.33
10	0	21.33
20	0.07	21.40
30	0.15	21.48
40	1.83	23.16

3 Heat Exchanger Network Design

3.1 Network from Aspen Energy Analyzer

The network was first made in Aspen Energy Analyzer with Recommend Designs, on the corrected stream data and with HP and MP steam raising as cold utilities. Design A_Design2 is taken. It is in Figure 3 and its numbers are in Table 7.

Table 7: Aspen Energy Analyzer network, design A_Design2.

Item	Value
Heat exchangers	17
Shells	20
Total area	3834 m^2
Hot utility	0 MW
Cooling water	16.63 MW
MP steam raised	4.70 MW
Smallest approach	9.4 K, on ARCOND and PRODCOOL

The actual network area of 3834 m^2 is below the area target of 4389 m^2 because the network design uses individual heat transfer coefficients for each match rather than a global uniform value.

All ten streams reach their target temperatures, so the design meets the energy target exactly. The reactor heat goes to the S1 feed (preheat, vaporising and superheat), to both reboilers of COL-1 and COL-2, and the balance raises MP steam. RXOUT gives heat to the COL-2 reboiler, to the S1 preheat and to the stripper, and then goes to cooling water. The COL-2 condenser drives the stripper reboiler in three exchangers. ARCOND and PRODCOOL run 0.6 K below the 10 K approach on the cooling water side, which is why Aspen marks them.

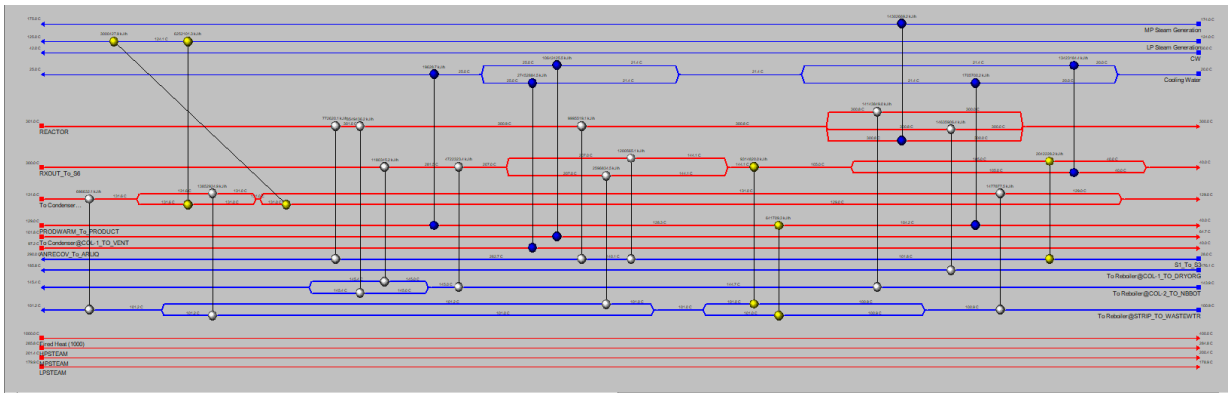


Figure 3: Heat exchanger network A_Design2 from Aspen Energy Analyzer, $\Delta T_{min} = 10$ K. Red lines are hot streams, blue lines are cold streams and utilities. Duties in kJ/h.

3.2 Steam level on the reactor

In A_Design2 the reactor is a source at 300°C, and it vaporises the feed at 262°C. In the plant this heat is carried by the coolant on the reactor shell, so the coolant would have to be above 273°C. Report 4 showed that the 25 mm bed peaks at 390.7°C against a 400°C limit, with its outlet near 240°C. The network assumes the 300°C isothermal outlet for heat recovery, which fixes the FEHE and the top of E-104. A coolant at 286°C, which is 70 bar steam, pushes the peak above the limit.

So 40 bar steam is taken on the shell, saturated at 250.4°C. The mean driving force against the 300°C bed is then about 50 K. The 9700 tubes of 25 mm give 1524 m^2 inside area, so the overall coefficient needed is

$$U = \frac{15.93 \times 10^6}{1524 \times 50} = 209 \text{ W/m}^2\text{K}, \tag{1}$$

which is a normal value for a packed-bed tube wall.

So the reactor works at an effective ΔT of about 50 K and not 10 K. From Table 6 this costs about 1.8 MW of hot utility, which is the vaporisation of the feed at 262°C.

3.3 Final network

The final network keeps the idea of A_Design2 but carries the reactor heat through the 40 bar steam header. The vaporiser stays on HP steam. Some small matches of A_Design2 are merged, so the unit count comes down from 17 to 14. The exchangers are in Table 8. Areas are from $Q = UA\,\Delta T_{lm}$, with U from the typical ranges in Sinnott and Towler [6]. We tried to put this network into Aspen Energy Analyzer also, as a second design, but it did not close in the time we had. So its balance is checked by hand, in Table 9.

Table 8: Final heat exchanger network. New units are marked N.

Unit	Hot side	Cold side	Duty (MW)	U (W/m^{2}K)	Area (m^2)
E-101 (N)	REACT bed, 300°C	BFW to 40 bar steam	15.93	209	1524
E-102A (N)	40 bar steam	S1, 28.0 to 240.4°C	1.88	500	54.8
VAP-1	70 bar steam	S1, 240.4 to 262.7°C	1.80	500/900	87.6
FEHE	RXOUT, 300.0 to 288.2°C	S4, 262.7 to 290.0°C	0.21	100	129.8
E-104 (N)	RXOUT, 288.2 to 200.0°C	BFW to 10 bar steam	2.17	120	345.4
COOL-1	RXOUT, 200.0 to 120.0°C	Cooling water	1.97	150	108.7
COOL-2	RXOUT, 120.0 to 40.0°C	Cooling water	5.24	300	527.5
E-103 (N)	COL-2 overhead, 131.6 to 129.0°C	STRIP bottoms, 100.9 to 101.2°C	7.02	800	300.3
STRIP reb.	10 bar steam	STRIP bottoms	0.92	1500	7.8
COL-1 reb.	16 bar steam, let down from 40 bar	COL-1 bottoms	4.06	1000	202.4
COL-2 reb.	10 bar steam, let down from 40 bar	COL-2 bottoms	6.56	1000	186.0
ARCOND	ANRECOV, 97.2 to 40°C	Cooling water	7.63	EDR	147.0
COL-1 cond.	COL-1 overhead	Cooling water	2.93	600	105.9
PRODCOOL	Product, 129 to 40°C	Cooling water	0.66	350	52.8

E-101 is the reactor shell itself, so it needs only a steam drum and no extra tube area. E-103 replaces the COL-2 condenser. The COL-2 overhead at 0.2 bar condenses on the shell side of the stripper reboiler, with 28 K driving force. The vapour line at 0.2 bar is large, so the stripper is placed next to COL-2. VAP-1 is now only the vaporiser and E-102A preheats its feed. COOL-1 becomes a trim cooler after E-104.

3.4 Steam balance and comparison with target

Table 9: Steam balance of the integrated plant, MW.

40 bar header		10 bar header	
Raised in E-101	15.93	Raised in E-104	2.17
To E-102A	−1.88	To STRIP reboiler	−0.92
To COL-1 reboiler, via 16 bar	−4.06		
To COL-2 reboiler, via 10 bar	−6.56		
Exported	3.43	Exported	1.25

These balances assume the hot condensate is returned to the drum.

Purchased heating is 1.80 MW against zero in A_Design2, and 1.83 MW in the 40 K target of Table 6. Cooling water plus exported steam is 23.11 MW against 21.33 MW in A_Design2. Both gaps come mainly from the reactor steam level of Section 3.2. The final network has 14 units against 17 in A_Design2.

4 Utility Costs

Utility prices are those in the Aspen file from Report 3. Exported steam is credited at half its price, as a buyer has to be found for it. Boiler feed water is charged at USD 0.5 per tonne on all steam raised. For a fair base case, REACT and COOL-1 are put on cooling water, since they had no utility in Report 4. The comparison is in Table 10.

Table 10: Utility use and cost before and after heat integration, 8200 h/y.

	Before		After	
Utility, price	kg/h	USD/h	kg/h	USD/h
HP steam 70 bar, 0.035 USD/kg	8,792	307.7	4,307	150.7
MP steam 16 bar, 0.030 USD/kg	7,565	226.9	0	0
LP steam 10 bar, 0.026 USD/kg	25,923	674.0	0	0
Cooling water, 2×10^{-5} USD/kg	3,137,336	62.7	1,327,277	26.5
Boiler feed water, 0.0005 USD/kg	–	–	27,628	13.8
40 bar steam exported, 0.016 USD/kg	–	–	−5,235	−83.8
10 bar steam exported, 0.013 USD/kg	–	–	−1,932	−25.1
Total, USD/h		1271.4		82.2
Total, USD million/y		10.43		0.67
Total, Rs. crore/y		99.0		6.4

The saving is USD 9.75 million per year. Cooling water circulation drops to 1327 t/h, so the pump and the tower fans of Report 3 come down from 311 kW to 262 kW. With COMP at 213 kW and the 20 per cent allowance of Report 3, plant demand becomes about 570 kW, against 757 kW from the gas turbine. The turbine still burns 7.71 kmol/h of natural gas, which is 5.86 MMBtu/h, or USD 0.48 million per year at USD 10 per MMBtu.

5 Capital Cost

5.1 Method

Equipment is costed with the bare-module correlations of Turton et al. [18], on carbon steel, with pressure factors where needed. The base CEPCI is 397 (2001), brought to 797.9 (2023). Fixed capital is the grass-roots cost, $1.18\sum C_{BM} + 0.50\sum C^{0}_{BM}$. Items which do not fit the correlations are taken as installed packages at the costs stated in Table 12. Working capital is 15 per cent of the fixed capital.

The columns were not sized in Reports 3 and 4, so they are sized here. Table 11 gives the result, from Fair flooding with $C = 0.09$ m/s at 80 per cent of flood, 0.6 m tray spacing and 70 per cent tray efficiency on the Aspen stages. COL-2 is wide because it runs at 0.2 bar.

Table 11: Column sizes used for costing.

Column	Max vapour (kg/h)	Pressure (bar)	Diameter (m)	Sieve trays	Height (m)
COL-1	28,770	1.1	1.85	15	13.0
COL-2	48,800	0.2	3.50	33	24.0
STRIP	13,000	1.0	1.75	10	10.0

5.2 Result

Table 12: Capital cost of the integrated plant, 2023 USD.

Item	Size or basis	C_{BM} (USD million)
REACT	1828 m^2 (outside area), 40 bar shell, 9700 tubes of 25 mm	1.826
Steam drum	1.5 m × 4.5 m, 40 bar	0.207
COL-1	Table 11	0.297
COL-2	Table 11, vacuum	1.741
STRIP	Table 11	0.227
FLASH and DEC	Report 4 sizes	0.096
Carbon adsorbers, two	1.2 m × 8.2 m	0.176
Heat exchangers, 13 units	Table 8, floating head	2.990
COMP with motor	213 kW, centrifugal	0.706
Pumps	P-1, P-2 and two cooling-water pumps	0.320
Nitrobenzene storage	2 × 2300 m^3, two weeks	0.622
Aniline storage	2 × 1900 m^3, two weeks	0.559
Cooling tower	369 L/s, Towler correlation [6], installed	1.426
Gas turbine package	757 kW at USD 1500/kW, assumed	1.135
Thermal oxidiser package	2.6 MW, assumed	0.600
BFW treatment and deaerator	28 t/h, assumed	0.300
Catalyst first charge	11,430 kg at USD 20/kg	0.229
Carbon first fill	6800 kg at USD 2/kg	0.014
Total bare module, $\sum C_{BM}$		13.47
Fixed capital, grass roots		18.22
Working capital, 15 %		2.73
Total capital investment		21.00
	Rs. crore	199.0

The heat integration adds USD 0.56 million in exchangers and USD 0.21 million for the steam drum. The smaller cooling tower saves USD 1.07 million. So the integrated plant costs USD 0.30 million less in bare-module cost and USD 0.41 million less in fixed capital than the plant without integration, whose fixed capital is USD 18.63 million. This is a study estimate, accurate to about ±30 per cent.

As a check, the purchased cost of the same equipment is USD 6.68 million. With the Lang factor of 4.74 for a fluid plant [6], and the catalyst and carbon added, the fixed capital comes to USD 31.9 million. This is higher than the bare-module figure, because the Lang factor carries more for piping, offsites and indirect costs. The bare-module value is kept as the base case, and the Lang value is run as a sensitivity case in Table 17.

6 Techno-Economic Analysis

6.1 Basis

Table 13: Economic basis.

Item	Value
Operating time	8200 h/y, as in Report 4
Product	99,924 t/y at 99.45 wt% aniline
Aniline price	USD 1632/t, Rs. 155,000/t, Report 1
Nitrobenzene price	USD 900/t, term contract, Report 1
Hydrogen price	USD 1947/t, on-site SMR cost, Report 1
Natural gas	USD 10/MMBtu, middle of the Report 1 range
Construction	2 years, 60 and 40 per cent of fixed capital
Plant life	15 years after start-up
Depreciation	Straight line over 10 years, no salvage
Tax	25 per cent
Discount rate	12 per cent
Working capital	15 per cent of fixed capital, recovered at the end

6.2 Cost of production

Fixed costs follow the factors of Sinnott and Towler [6]. Labour is 15 operators, which is three per shift on five shifts, at USD 9000 per year each.

Table 14: Cost of production at full capacity.

Item	USD million/y	USD/t product
Nitrobenzene, 142,624 t/y	128.36	1284.6
Hydrogen, 7434 t/y	14.48	144.9
Utilities, Table 10	0.67	6.7
Natural gas to gas turbine	0.48	4.8
Catalyst, 3-year life	0.08	0.8
Activated carbon, 30-day cycle	0.08	0.8
Wastewater treatment, USD 5/m^3	0.18	1.8
Variable cost	144.33	1444.4
Labour, supervision and salary overhead	0.25	2.5
Maintenance, 3 % of fixed capital	0.55	5.5
Insurance and local taxes, 1 %	0.18	1.8
General plant overhead	0.52	5.2
Cash cost of production	145.83	1459.4
Revenue	163.03	1631.6

The wastewater charge is high on purpose. It covers the cyclohexylamine in WASTEWTR, which carbon does not hold, so the water goes to biological treatment.

6.3 Profitability

Table 15: Profitability, base case.

Item	Value
Gross profit	USD 17.21 million/y
Depreciation	USD 1.82 million/y
Profit before tax	USD 15.38 million/y
Profit after tax	USD 11.54 million/y
Annual cash flow, years 1 to 10	USD 13.36 million/y
NPV at 12 per cent	USD 61.3 million (Rs. 583 crore)
IRR	50.4 per cent
Simple payback on total capital	1.6 years
Return on investment, profit after tax / total capital	55 per cent

The year-wise cash flow is in Table 16. Year −1 and year 0 are construction. Working capital goes in at year 0 and comes back in year 15. The tax shield of depreciation ends after year 10, so the cash flow drops slightly from year 11.

Table 16: Cash flow, USD million. Discounted at 12 per cent.

Year	Cash flow	Cumulative	Cumulative, discounted
−1	−10.93	−10.93	−10.93
0	−10.02	−20.95	−19.88
1	13.36	−7.59	−9.23
2	13.36	5.77	0.28
3	13.36	19.13	8.77
4	13.36	32.49	16.35
5	13.36	45.85	23.12
6	13.36	59.21	29.17
7	13.36	72.57	34.56
8	13.36	85.93	39.38
9	13.36	99.29	43.68
10	13.36	112.65	47.52
11	12.90	125.55	50.83
12	12.90	138.45	53.79
13	12.90	151.36	56.43
14	12.90	164.26	58.79
15	15.64	179.90	61.34

The discounted cash flow turns positive in year 2 after start-up.

6.4 Sensitivity

The base aniline price of USD 1632/t corresponds to the Rs. 155,000/t target from Report 1.

Table 17: Sensitivity of the result. One change at a time from the base case.

Case	NPV (USD million)	IRR (%)	Payback (y)
Base case	61.3	50.4	1.6
Aniline USD 1600/t, top of historical quoted band	47.0	42.7	1.9
Aniline USD 1550/t, off-spec price-cut	19.3	26.5	3.1
Aniline USD 1500/t, bottom of quoted band	1.4	13.2	6.0
Nitrobenzene USD 960/t, spot price	22.3	28.2	3.0
Fixed capital +30 per cent	54.7	40.0	2.1
Fixed capital +50 per cent	50.3	35.0	2.4
Fixed capital by Lang factor, USD 31.9 million	44.7	30.2	2.8
Discount rate 15 per cent	48.3	50.4	1.6
No heat integration	16.4	24.0	3.6

The project is driven by the aniline and nitrobenzene prices, not by capital. NPV goes to zero at an aniline price of USD 1497/t or a nitrobenzene price of USD 994/t. A 50 per cent rise in capital costs only USD 11 million of NPV, and even the Lang capital keeps the IRR at 30 per cent. Heat integration is worth USD 45 million of NPV, because it takes USD 9.75 million a year off the utility bill.

7 Objectives Which Could Not Be Accomplished, With Reasons

Both objectives of this report, the network design by pinch technology and the utility and techno-economic study, are complete. The product and discharge specifications set in Report 1 are separate from these objectives and are carried forward. The position on each is below.

Aniline purity. 99.45 wt% against a target of 99.9 wt%, unchanged from Report 4. Cyclohexylamine at 0.48 wt% holds it back. A light-ends draw at the top of COL-2 is the route. The product is priced at the full aniline price in Section 6, which assumes the purity is met.

Nitrobenzene in the product. 313 ppm against a target of 5 ppm. A taller COL-2 is the route, as in Report 4. Ten more trays on the 3.5 m vacuum column cost about USD 0.4 million, which does not change the NPV much.

Aniline in the wastewater. 216 ppm into the carbon bed against a discharge limit of 1 ppm. The bed holds it below 1 ppm on the design basis of Report 4. A breakthrough curve is still needed to confirm this.

Cyclohexylamine in the wastewater. 4.28 wt%. It is now costed as biological treatment at USD 5/m^3, Section 6.

Second decanter on WATOVHD. Still open loop, because WATOVHD is the only exit benzene has. A benzene column is needed first.

Purchased heating. 1.80 MW of 70 bar steam for VAP-1, against zero in the Aspen network. It comes from the 40 bar steam level that the bed limit sets. A thermal-oil loop on the first part of the bed, where most heat is released, could supply it, and that is the route.

Reaction kinetics. The two side-reaction rate constants are still fitted, not regressed.

8 Any Other Challenges

REACT read as a cold stream. The Energy Analysis took the reactor as a stream heated from 160 to 300°C by fired heat. This is because RPlug has no utility option and its outlet is hotter than its inlet. The first target was 27 MW of heating, which is wrong. The stream editor of Aspen Energy Analyzer does not allow a cold stream to be changed to hot, so the stream was deleted and re-entered as a hot stream. This had to be done twice, as the first corrected file was not saved.

Scope of the analysis. The Define Scope option works only on whole flowsheets, not on single blocks. So DEC and FLASH were removed in the stream table and not in the scope.

Automatic network design. On the first run Recommend Designs gave ten networks, and all had infeasible exchangers. After the reactor stream was corrected and fired heat was replaced by HP and MP steam raising, the second run gave A_Design2 with all streams satisfied. We also tried to put our own 14-unit network into Aspen as a second design, but it did not close in the time we had, so that network is checked by hand.

9 Number of Hours Spent During This Period

Approximately 5 hours per member, which is 30 hours for the group.

10 Contributions From Individual Members

Nithin T M and Aditya Nitin Patil — the Energy Analysis in Aspen Plus, the correction of the reactor stream, the targets, the reactor steam level and the final network, and the manual PTA calculation compilation of this report.

Nithin D H and Shubham Singh — the utility prices and steam balance, the column sizing, the vessel and storage costs, and the gas turbine fuel cost.

Shivam Dongare and Sajag Masane — the network A_Design2 in Aspen Energy Analyzer with its curves and tables, the exchanger areas and costs, and the cash flow and sensitivity cases.

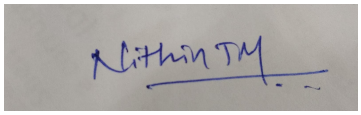
All six members have reviewed the complete report.

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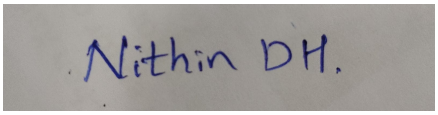
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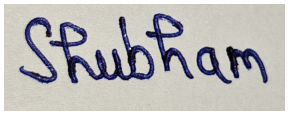
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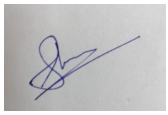
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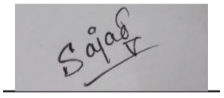
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